Hammad Amer

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Dynamic Software Engineer at the intersection of Artificial Intelligence and Full-Stack Development, experienced in deploying PyTorch models within scalable MERN architectures. Skilled in building end-to-end products, from multi-agent ML pipelines to robust, user-facing web and mobile applications.

EDUCATION

FAST NUCES | *Islamabad* **Aug 2022 - Jun 2026**

Bachelor of Science in Computer Science

Coursework: Data Structures & Algorithms, Generative AI, MLOps, Cloud Computing, Web Programming, Information Security

TECHNOLOGIES

Core Development: React, Node.js, Express.js, Python, C++, Java, Kotlin, MongoDB, MySQL, Firebase, MPI, OpenCL
Cloud & Infrastructure: AWS, Microsoft Azure, Kubernetes, Docker, GitHub Actions, Linux
AI & MLOps: PyTorch, TensorFlow, MLflow, Airflow
Engineering Practices: Agile, CI/CD, Test Driven Development (TDD)

WORK EXPERIENCE

Code Generation | AI/ML Developer Intern **Jun 2025 – Sep 2025**

- Architected and developed an automated Financial Data Parser leveraging multi-agent architectures and genetic algorithms, transitioning complex financial datasets into structured, actionable formats for the reconciliation system.
- Owned the end-to-end integration of these machine learning models into the startup's software applications, collaborating closely with the engineering team to ensure seamless deployment and reliable performance.
- Streamlined data processing workflows by delivering a production-ready parsing solution, demonstrating the ability to rapidly design and ship complex AI features within a fast-paced environment.

Freelance Technical Consultant | Fiverr **May 2023 – Feb 2024**

- Provided freelance technical consulting services, improving website performance and search visibility through infrastructure audits, technical SEO optimizations, and performance tuning.

PROJECTS

SpineScan (Final Year Project) – AI-powered Lumbar Spine MRI Diagnostic Tool | **Stack:** PyTorch, YOLOX, MERN
GitHub: *github.com/Hammad-Amer/SpineScan*

- Designed a multi-stage deep learning pipeline using YOLOX detection and timm backbones to classify 25 spinal conditions from DICOM MRI scans, achieving 94% multi-label accuracy.
- Built a full-stack MERN dashboard with Grad-CAM explainability, single-study DICOM inference, and an optimized GPU training pipeline using FP16 mixed precision and gradient accumulation, reducing training time by 60% compared to baseline.

Multimodal RAG & Semantic Search System | Stack: LLaMA2, Sentence-BERT, CLIP/BLIP, FAISS, LlamaIndex, Flask
GitHub: *github.com/Hammad-Amer/MultiModalRag*

- Built a RAG pipeline handling text and image queries from PDFs, integrating LLMs for context-aware answers with CoT/few-shot prompting.
- Deployed a web interface for querying and retrieval, showing ranked chunks, images, and generated responses.

ConnectMe – Real-time social media Android app | **Stack:** Kotlin, XML, Firebase, XAMPP, SQL, FastAPI, Flask, Agora API
GitHub: *github.com/Hammad-Amer/ConnectMe*

- Built a social platform with real-time messaging, audio/video calls, stories, posts, followers, and user profile management.
- Initially implemented using Firebase, later migrated to XAMPP/PHP for faster flows, and leveraged Flask/FastAPI to build a spam message classifier backend and handle APIs efficiently.

ACHIEVEMENTS & HONORS

Dean's List Honoree (2×) — FAST NUCES Islamabad